

Header Image: assets/header.png

[□□ English](README_SUMMARY.md) | [□□ Türkçe](README_SUMMARY_TR.md)

MertFormer Titan - External Summary (Build 30)

What This Project Is

MertFormer Titan is a mobile-first, offline-capable AI architecture designed for controlled, auditable, and human-supervised use.

It combines BitNet (1.58-bit), Liquid dynamics, and MoE routing under a production-first engineering approach.

Current Status

- **Stage**: Pilot-ready, pre-training baseline (`Build 30`)
- **Positioning**: Proof-of-system for real-world constrained environments
- **Not claimed yet**: Final benchmark superiority and production-grade performance claims (require trained checkpoint evidence)

Safety and Governance Policy

- Human-in-the-loop is mandatory for operational decisions.
- Audit trail and policy boundaries are mandatory on orchestrator/runtime side.
- Unauthorized surveillance, hidden tracking, and non-consensual intervention are out of scope.
- Security and governance gates must pass before pilot performance claims.

Verified Local Evidence (Latest Run)

```
| Gate | Result |
| :--- | :--- |
| `python3 -m pytest -q` | `59 passed, 3 skipped` |
| `.titan-venv/bin/python -m ruff check .` | `All checks passed` |
| `bash scripts/verify_all.sh` | `[verify] OK` |
```

Closure artifacts:

- `reports/closure\_57\_matrix.json`
- `reports/closure\_57\_matrix.md`
- `reports/closure\_57\_matrix\_TR.md`

Quick Start (External Reviewer)

1. Create/refresh virtual environment:

```
bash scripts/bootstrap_venv.sh
```

2. Run full offline verification gate:

```
bash scripts/verify_all.sh
```

3. Check training readiness (strict gate):

```
bash run.sh --train-ready
```

4. Start training when compute + dataset prerequisites are satisfied:

```
TITAN_OFFLINE=0 TITAN_INSTALL=1 TITAN_PROFILE=stable bash run.sh
```

External Pilot Usage Model

- Run reproducible validation gates in the customer environment.
- Share machine-readable pilot evidence (logs/reports), not unverifiable claims.
- Use NDA/private data-room flow for sensitive technical details.
- Keep measured vs projected results clearly separated.

Claim Boundary (Important)

- Before a fully trained checkpoint and repeatable benchmark outputs, this repo remains:

- ****Pilot-ready engineering baseline****
- ****NOT ELIGIBLE FOR FINAL BENCHMARK CLAIMS****

Useful Docs

- Main docs: README.md, README_TR.md
- Usage guide: USAGE_GUIDE.md,
USAGE_GUIDE_TR.md
- SDK guide: SDK_GUIDE.md, SDK_GUIDE_TR.md
- Safety/policy: SECURITY.md, USE_POLICY.md